

Can LLMs Generate Novel Research Ideas?

A Large-Scale Human Study with 100+ NLP Researchers

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Abstract

Recent advancements in large language models (LLMs) have sparked optimism about their potential to accelerate scientific discovery, with a growing number of works proposing research agents that autonomously generate and validate new ideas. Despite this, no evaluations have shown that LLM systems can take the very first step of producing novel, expert-level ideas, let alone perform the entire research process. We address this by establishing an experimental design that evaluates research idea generation while controlling for confounders and performs the first head-to-head comparison between expert NLP researchers and an LLM ideation agent. By recruiting over 100 NLP researchers to write novel ideas and blind reviews of both LLM and human ideas, we obtain the first statistically significant conclusion on current LLM capabilities for research ideation: we find LLM-generated ideas are judged as more novel ($p < 0.05$) than human expert ideas while being judged slightly weaker on feasibility. Studying our agent baselines closely, we identify open problems in building and evaluating research agents, including failures of LLM self-evaluation and their lack of diversity in generation. Finally, we acknowledge that human judgements of novelty can be difficult, even by experts, and propose an end-to-end study design which recruits researchers to execute these ideas into full projects, enabling us to study whether these novelty and feasibility judgements result in meaningful differences in research outcome.¹

1 Introduction

The rapid improvement of LLMs, especially in capabilities like knowledge and reasoning, has enabled many new applications in scientific tasks, such as solving challenging mathematical problems (Trinh et al., 2024), assisting scientists in writing proofs (Collins et al., 2024), retrieving related works (Ajith et al., 2024, Press et al., 2024), generating code to solve analytical or computational tasks (Huang et al., 2024, Tian et al., 2024), and discovering patterns in large text corpora (Lam et al., 2024, Zhong et al., 2023). While these are useful applications that can potentially increase the productivity of researchers, it remains an open question whether LLMs can take on the more creative and challenging parts of the research process.

We focus on this problem of measuring the *research ideation* capabilities of LLMs and ask: are current LLMs capable of generating novel ideas that are comparable to expert humans? Although ideation is only one part of the research process, this is a key question to answer, as it is the very first step to the scientific research process and serves as a litmus test for the possibility of autonomous research agents that create their own ideas. Evaluating expert-level capabilities of LLM systems is challenging (Bakhtin

¹Interested researchers can sign up for this end-to-end study at: <https://tinyurl.com/execution-study>. We release our agent implementation and all human review scores at: <https://github.com/NoviScl/AI-Researcher>.

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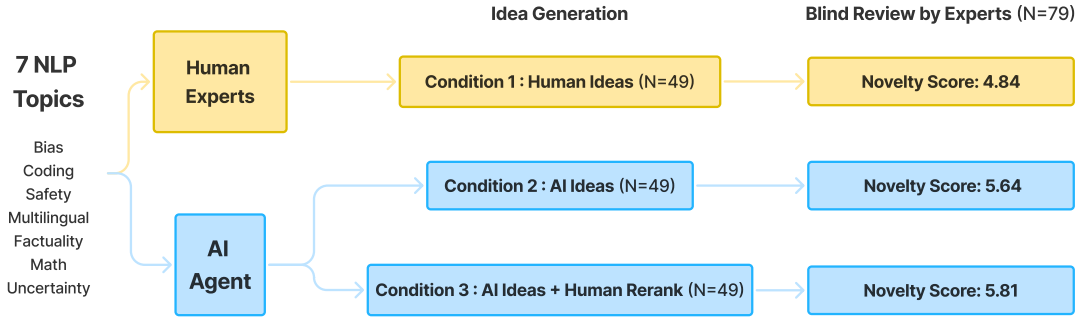


Figure 1: Overview of our study: we recruit 79 expert researchers to perform blind review of 49 ideas from each of the three conditions: expert-written ideas, AI-generated ideas, and AI-generated ideas reranked by a human expert. We standardize the format and style of ideas from all conditions before the blind review. We find AI ideas are judged as significantly more novel than human ideas ($p < 0.05$).

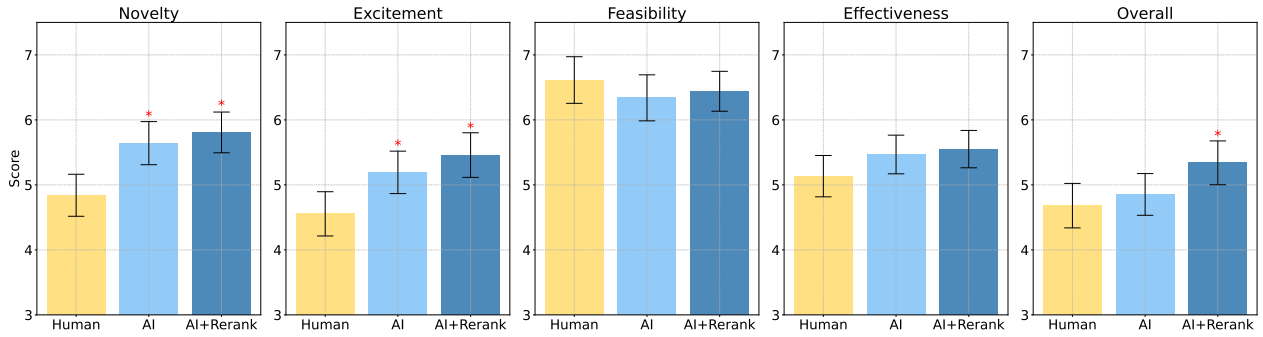


Figure 2: Comparison of the three experiment conditions across all review metrics. Red asterisks indicate that the condition is statistically better than the Human baseline with two-tailed Welch’s t-tests and Bonferroni correction. All scores are on a 1 to 10 scale. More detailed results are in Section 5.

et al., 2022, Collins et al., 2024), and research ideation takes this to an extreme. Qualified expert researchers are difficult to recruit at scale, evaluation criteria can be highly subjective, and it is difficult for even the best experts to judge the quality of an idea (Beygelzimer et al., 2021, Simsek et al., 2024). We address these challenges directly, recognizing that for important, high-stakes tasks like research ideation, there is no substitute for a large-scale expert evaluation. We design a carefully controlled comparison of human and LLM ideas that overcomes sample size and baseline problems present in earlier small-scale evaluation studies. Our study recruited a large pool of over 100 highly qualified NLP researchers to produce human baseline ideas and perform blind reviews of human and LLM ideas. To reduce the possibility that confounding variables affect our outcome measures, we enforce strict controls that standardize the styles of human and LLM ideas and match their topic distribution. We compare our human expert baseline with a simple and effective LLM agent that incorporates retrieval augmentation and adopts recent ideas in inference-time scaling, such as overgenerating and reranking LM outputs. These measures allow us to make statistically rigorous comparisons between human experts and state-of-the-art LLMs (Figure 1).

Our evaluation-centric approach complements many recent methods-centric works that attempt to instantiate research agents (Baek et al., 2024, Li et al., 2024, Lu et al., 2024, Wang et al., 2024, Yang et al., 2024). The majority of these works rely on fast and lower-cost evaluation surrogates – either by decreasing the number of expert reviewers (Baek et al., 2024, Li et al., 2024, Wang et al., 2024, Yang et al., 2024), constraining the length and detailedness of the ideas (Wang et al., 2024, Yang et al., 2024), or relying on LLM-as-a-judge (Lu et al., 2024). They do not perform the large-scale human comparison studies that are needed to answer the motivating question of our work. Our work takes the opposite approach, performing a year-long and high-cost evaluation that provides human expert baselines and a standardized evaluation protocol to serve as a foundation for future follow-up studies and methods work. Through nearly 300 reviews across all our conditions, we find that AI-generated ideas are judged as more novel than human expert ideas ($p < 0.05$), which holds robustly under multiple hypothesis correction and across different statistical tests. We find some signs that these gains are correlated with excitement and overall score, and may come at the slight expense of feasibility, but our study size did not have sufficient power to conclusively identify these effects (Figure 2).

Qualitative analysis of free-text responses in our review corroborates these findings on novelty and feasibility. Apart from evaluating the ideas, we also analyze the LLM agent, showing limitations and open problems – despite excitement about inference-time scaling of LLMs, we find that they lack idea diversity when we scale up idea generation, and they cannot currently serve as reliable evaluators.

2 Problem Setup

The central experiment of our work is a comparison of human- and LLM-generated ideas. While this goal is simple, there is no existing consensus on how to formulate the task of research ideation and evaluation, and we begin by defining the key aspects of our experiment design.

We think of research idea evaluation as consisting of three separate components: 1). the idea itself, generated in response to our instructions, 2). the writeup which communicates the idea, and 3). the evaluation of the writeup by experts. We outline our experiment design in each of these three parts with particular focus on potential confounders, such as the area of research, the format of a research idea, and the evaluation process.

Ideation Scope and Instructions Research ideas can take many different forms. They can be simple tricks to improve model performance, or they may be large-scale research programs that form the basis of a Ph.D. thesis. Any experiment on ideation must carefully balance the realisticness and interestingness of a research idea with the practical realities of eliciting ideas from a large population. In our case, these tradeoffs are even more pronounced, as we have designed our ideation experiments so that the resulting ideas can be executed by experts in a follow-up set of experiments.

These constraints have led us to study prompting-based NLP research as a testbed for our study. Prompting research has been popular in recent years of NLP and AI research (e.g., Chen et al., 2023, Diao et al., 2024, Madaan et al., 2023, Qin et al., 2024, Schulhoff et al., 2024, Si et al., 2023, Wang et al., 2023, Wei et al., 2022, Yao et al., 2023, Yasunaga et al., 2024, Zhou et al., 2023, inter alia). This class of projects strikes a reasonable trade-off among our constraints. The most impactful prompting projects like chain-of-thought have had a major influence on LLM performance (Wei et al., 2022), and prompting projects are executable with minimal computing hardware.

We further structure our ideation process to avoid selection-bias-based confounders in ideation. If we simply ask LLMs and humans to produce ideas on ‘prompting topics’, we may find that LLMs and humans differ in the types of research ideas they produce (for example, LLMs may naturally suggest more projects on safer topics, which might be judged as less exciting by humans). This would

lead us to simply measure misalignment in research topic preference between LLMs and humans, which is not the goal of our study. To address this possibility, we define a set of seven specific research topics extracted from the Call For Papers page of recent NLP conferences such as COLM.² Specifically, our topics include: Bias, Coding, Safety, Multilinguality, Factuality, Math, and Uncertainty (see Appendix A for a complete description of these topics).

Each human and LLM participant of the ideation experiment receives the same set of natural language instructions including the same topic description, idea template, and demonstration example to ensure a fair comparison. For human participants, we additionally allow them to select a preferred topic from the list, and for each selected topic, we generate a corresponding LLM idea. This exactly matches the idea topic distribution between the LLM and human participants, while ensuring that human experts are able to select topics according to their expertise.

Idea Writeup An idea can only be evaluated if it is written up to be communicated, but this writing process introduces many additional potential confounders. Human researchers may write in ways that subtly signal quality research, such as including more examples and implementation details. The format of the writeup functions as a way to scaffold what contents should be included and the level of detailedness. Ideally, we want both human and LLM participants to provide all the necessary implementation details for their generated ideas.

We take inspiration from guidelines used in grant submissions and introduce a template to specify the structure and detailedness of idea proposals. Specifically, we construct a template that includes fields for the title, problem statement, motivation, proposed method, step-by-step experiment plan, test case examples, and the fallback plan. Both the LLM agent and the human idea writers are instructed to follow this template and our provided demonstration examples to produce a project proposal as the output (see Appendix B for the full template and Appendix C for the demo example).

Even with these templates, there may be subtle writing style cues that affect the outcome measure. For example, humans may tend to write in a more engaging and informal tone. To reduce this possibility further, we developed a style normalization module that uses an LLM to convert all ideas into the same writing and formatting style without changing the original content. Our small-scale human study shows that such a normalization approach leads to a 50% accuracy for expert human judges who are asked to distinguish AI ideas from human ideas. Finally, the use of an LLM style anonymizer has the possibility of substantively changing the content of the ideas. To rule this out, the first author of this paper manually verified each human idea proposal to ensure all contents of the original ideas were preserved. We present the full prompt used in Appendix D.

Review and Evaluation Reviewing research ideas is notoriously subjective, so we want to design a review form that defines all review criteria clearly to standardize and anchor the evaluations as much as possible. At the same time, we want our review criteria and measured variables to capture all the desiderata of high-quality research ideas.

We follow best practices from AI conference reviewing (e.g., ICLR and ACL) when designing the review form, where we define four breakdown metrics including novelty, excitement, feasibility, and expected effectiveness, apart from the overall score. For each metric, we ask for a numerical score on a 1-10 scale along with a free-text rationale. We provide clear definitions and grounding for each numerical scale to calibrate all reviewers’ standards (see Appendix E for the full review form).

Our blind review evaluation will compare ideas from three different conditions:

1. Human Ideas: Idea proposals written by our recruited expert researchers.

²<https://colmweb.org/cfp.html>

2. **AI Ideas:** Idea proposals generated by our LLM agent. We directly take the top-ranked ideas from the agent’s output.
3. **AI Ideas + Human Rerank:** Idea proposals generated by our LLM agent. The first author of this paper manually selected the top-ranked ideas out of all the LLM agent’s generations rather than relying on the LLM ranker in order to better estimate the upper-bound quality of AI ideas.

In the next two sections, we instantiate how our LLM agent generates ideas and how our expert participants generate and review the ideas.

3 Idea Generation Agent

We build a simple but effective LLM ideation agent to compare with the human expert baseline. Rather than focusing on innovating the agent itself, we adhere to a minimalist design principle, aiming to understand the current capabilities of LLMs in idea generation. Our research ideation agent has three essential components: paper retrieval, idea generation, and idea ranking, which we will describe in detail below.

3.1 Paper Retrieval for RAG

To ground idea generation, the agent needs to retrieve papers related to the given research topic, so that it will be aware of related works when generating new ideas. To do so, we leverage retrieval-augmented generation (RAG), which has demonstrated effectiveness on many knowledge-intensive tasks (Lewis et al., 2020, Shi et al., 2024). Concretely, given a research topic (e.g., “novel prompting methods that can improve factuality and reduce hallucination of large language models”), we prompt an LLM to generate a sequence of function calls to the Semantic Scholar API. We use `claude-3-5-sonnet-20240620` as the backbone model for our agent but the pipeline should generalize to other LLMs as well. The paper retrieval action space includes: `{KeywordQuery(keywords), PaperQuery(paperId), GetReferences(paperId)}`. Each action generation is grounded on the previous actions and executed results. We keep the top $k=20$ papers from each executed function call and stop the action generation when a max of $N=120$ papers have been retrieved. We then use the LLM to score and rerank all retrieved papers based on three criteria: 1) the paper should be directly relevant to the specified topic; 2) the paper should be an empirical paper involving computational experiments;³ 3) the paper is interesting and can inspire new projects. The LLM is prompted to score each retrieved paper on a scale of 1 to 10 based on these criteria and we use the top-ranked papers for the next step of idea generation.

3.2 Idea Generation

Our key insight for idea generation is to generate as many candidate ideas as possible. Our intuition is that only a small fraction of all generated ideas might be high-quality, and we should be willing to expend inference-time compute to generate more candidates so that we can later use a reranker to discover the “diamond in the rough”. This aligns with existing results showing that scaling inference compute with repeated sampling can boost LLM performance on various coding and reasoning tasks (Brown et al., 2024, Li et al., 2022). Specifically, we prompt the LLM to generate 4000 seed ideas on each research topic. The idea generation prompt includes the demonstration examples and the retrieved papers. We craft $k=6$ demonstration examples by manually summarizing exemplar

³Note that we exclude position papers, survey papers, and analysis papers throughout this study since their evaluation tends to be very subjective.